Reviewer Pack — Minimal Order of Lie Algebroid Dimensions and Circuit Lower …

Entry 32674bf7a448 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Lie Algebroid Dimensions and Circuit Lower Bounds for k-CLIQUE Inclusion

Entry ID: `32674bf7a448` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-06 11:25:29 UTC

1. Conjecture

Field A (mathematical branch): Lie Algebroids Theory **Field B** (complexity object): Boolean Circuit Complexity (Circuit Lower Bounds for k-CLIQUE)

Statement:

> For every CNF formula φ representing a k-CNF, the minimal dimension of its corresponding Lie algebroid L(φ) is O(k^(2/3)) in terms of the size n of φ.

Rationale (proposer's reasoning):

> Lie algebroids provide an algebraic framework that can encode the complexity of certain computational problems. The conjecture suggests that this encoding could be useful for understanding circuit lower bounds for k-CLIQUE, which is a fundamental problem in boolean circuit complexity.

Taxonomy category: `LieAlgebroids` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `89d191e687852251`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> For a given CNF formula φ, if the dimension of its Lie algebroid L(φ) is at least O(k^(2/3)) * n^(2/3), where n is the size of φ and k is the clique size, then the conjecture is supported.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM₁	LLM₂
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `ADJACENT_OK` against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `('Lie Algebroids' AND 'circuit complexity')` - `('k-CLIQUE inclusion' AND 'dimension of Lie algebras')` - `('minimal order of dimensions' AND 'Boolean circuits lower bounds')`

Top relevant hits considered: - [http://arxiv.org/abs/1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [http://arxiv.org/abs/1311.2475v5] On almost complex Lie algebroids - [http://arxiv.org/abs/1401.3795v2] Relationship between Nichols braided Lie algebras and Nichols algebras - [http://arxiv.org/abs/2503.21515v1] Discrete inclusions from Cuntz-Pimsner algebras - [http://arxiv.org/abs/0708.4200v2] Braided-Lie bialgebras associated to Kac-Moody algebras - [http://arxiv.org/abs/0708.1267v1] Borel subalgebras of root-reductive Lie algebras - [http://arxiv.org/abs/1605.04207v1] Functional lower bounds for arithmetic circuits and connections to boolean circuit complexity - [http://arxiv.org/abs/1410.4915v3] Rankin-Selberg L-functions in cyclotomic towers, III

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def tseitin_encoding(variables, clauses):
        new_vars = {}
        tseitin_clauses = []

        for clause in clauses:
            if len(clause) == 1:
                l = clause[0]
                if l < 0:
                    l = -l
                if l not in new_vars:
                    new_var = max(new_vars.values()) + 1 if new_vars else 1
                    new_vars[l] = new_var
                tseitin_clauses.append([new_var, -l])
            else:
                new_var = max(new_vars.values()) + 1 if new_vars else 1
                new_vars[new_var] = new_var
                for l in clause:
                    if l < 0:
                        l = -l
                    tseitin_clauses.append([new_var, -l])
```

```
                tseitin_clauses.append([-new_var] + [-l for l in clause])

        return list(new_vars.values()), tseitin_clauses

    def lie_algebroid_dimension(n):
        # Placeholder function to compute the Lie algebroid dimension
        return n ** (2/3)

    def k_clique_circuit_size(k, n):
        # Placeholder function to compute the circuit size for k-CLIQUE
        return k * n

    sizes = [5, 10, 15, 20, 30, 40]
    total_metric_value = 0
    instances_tested = 0
    conjecture_holds = True
    counterexample = ""

    for size in sizes:
        n = random.randint(1, size)
        k = random.randint(1, min(n, 5))

        variables = list(range(1, n + 1))
        clauses = []
        for _ in range(k):
            clause = [random.choice(variables) for _ in range(random.randint(1, n))]
            clauses.append(clause)

        literals, tseitin_clauses = tseitin_encoding(variables, clauses)
        lie_dim = lie_algebroid_dimension(n)
        circuit_size = k_clique_circuit_size(k, n)

        if lie_dim < Fraction(k ** (2/3) * n ** (2/3)):
            conjecture_holds = False
            counterexample = f"n={n}, k={k}, Lie dim={lie_dim}, Circuit size={circuit_size}"
            break

        total_metric_value += lie_dim
        instances_tested += 1

    return {
        "metric_name": "Lie algebroid dimension",
        "metric_value": total_metric_value / instances_tested,
        "instances_tested": instances_tested,
        "n_max": max(sizes),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
```

```
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fract
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fract
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b36b6728.py", line 96, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b36b6728.py", line 83, in run_trial
    "metric_value": total_metric_value / instances_tested,
                    ~~~~~~~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test crashed before producing any data, which means we cannot verify if the dimension of the Lie algebroid meets the support condition for the conjecture. | next: Re-run the test without crashing to verify the dimension of the Lie algebroid against the conjectured bound.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12716
2	propose	ollama_remote	glm4:latest	0	0	12139

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
3	preregistration	ollama_remote	glm4:latest	0	0	10416
4	novelty	ollama_remote	glm4:latest	0	0	13167
5	novelty	ollama_remote	glm4:latest	0	0	9655
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14631
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9713
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9449
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10851
10	judge	ollama_remote	glm4:latest	0	0	13047

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 115785 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/32674bf7a448.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/32674bf7a448.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/32674bf7a448.tar.gz` (if generated)